

SD7003 Response to Discrete Vortical Gust Encounter

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The generation and evolution of a discrete vortex was simulated using unsteady panel method and the results were compared with PIV measurements in a field of view 5 chords downstream of the gust generator. The dynamics of the discrete vortex gust encounter of an SD7003 wing were studied across a span of angles of attack. Discrete vortical gusts with different strengths and trajectories were generated and characterized using Time Resolved Particle Image Velocimetry (TR-PIV) in the University of Dayton Water Tunnel (UD-Wat) using a pitch-ramp-return motion. Simultaneous force and TR-PIV measurements were taken of the gust encounter of the SD7003 wing. The wing’s response to each gust profile was studied across a span of angles of attack. The impact of angle of attack and vortex offset distance impacts the C_L recovery of the wing during the gust exit.

Nomenclature

A	=	Oscillation amplitude (degrees)	S	=	Shear (1/s)
u	=	x component of velocity (m/s)	C_L	=	Coefficient of Lift
U_∞	=	Free stream velocity (m/s)	Abbreviations		
v	=	y component of velocity (m/s)	FOV	=	Field of View
t^*	=	Convective time	TR-PIV	=	Time Resolved Particle Image
$\dot{\alpha}$	=	Pitch Rate (degrees/s)	Velocimetry		
$\dot{\alpha}_N$	=	Normalized Pitch Rate	DVM	=	Discrete Vortex Method
Q	=	Q-Criterion (1/s ²)	UD-WaT	=	University of Dayton Water Tunnel
ω	=	Vorticity (rad/s)			

I. Introduction

As unmanned aerial vehicles have found their way into small, low-speed air vehicles are beginning to expand into many sectors including agriculture, defense, and recreation. These aerial vehicles fly under vastly different conditions when compared to traditional aircraft. These new vehicles fly at slower speeds and closer to the ground than most traditional aircraft. A light unmanned aircraft flying close to the ground will be greatly impacted by unsteady aerodynamic disturbances. [1] These aerodynamic disturbances can be categorized as vortical, transverse, and streamwise gusts. This study will focus on the generation and characterization of a ramp-hold-return gust profile. Vortical gusts can be generated in pairs with a periodic profile, and a discrete gust can be generated by rapidly pitching a plate once. [2] While a rapidly pitched plate will create a discrete gust, the subsequent shedding off the pitched plate creates large disturbances to the flow following the gust. [3] This study will focus on a ramp-hold-return profile which consists of a rapid pitching motion, a short delay, and a significantly slower return rate. The initial pitch creates the discrete gust, and the slower return rate is intended to reduce the shedding from the pitched plate while moving slow enough to prevent a counter rotating vortex.

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Several studies have shown that rapidly pitching a symmetric airfoil can create vortical gusts of varying strength. Vadher and Babinsky [4] rapidly pitched a symmetric airfoil through a tow tank to investigate the gust. A focus of this study is to minimize the secondary effects of the gust generation method. As such, a flat plate gust generator will be used to reduce the shear layer shedding off the upstream gust generator. Vadher and Babinsky showed that a slower starting pitch rate results in a secondary vortex forming and an overly rapid pitching motion prevents the formation of a coherent vortex. [5]

A study by Killian et. al [6] studied the mitigation of a periodic gust profile using a closed loop controller. The gusts investigated in this study were less coherent than the discrete gusts tested in this study, but the strength of vorticity and the effects on C_L are similar. A previous study in the UD-WAT investigated a flat plate wing's response to a series of different strength vortical gusts generated from a pitch-hold gust profile. [7] The initial gust encounter was well characterized, but the gust was followed by a region of significant shedding from the pitched gust generator upstream of the field of interest. The flat plate wing was kept at an angle of attack of 0. A closed loop PID controller was employed in order to mitigate the influence the gust had on the C_L of the wing. The study identified a strong wake following the gust encounter. The gust was followed by a gap of relaxed flow before the vorticity of the gust generator wake entered the field of view and affected the wing. The strength and irregularity of the wake proved difficult to control with the PID system.

In this study, the generation and evolution of a vortical gust will be simulated using discrete vortex method solution. The DVM simulation will be compared to PIV measurements in the University of Dayton Water Tunnel. The size, strength, and trajectory of each gust profile will be measured and compared to the simulation results. A wing with a SD7003 airfoil will be placed 5 chords downstream from the gust generator in the field of view from the gust characterization. Simultaneous force and PIV measurements will be conducted during the gust encounter. The SD7003 wing is expected to have an extended, shallow stall according to the study by Ol and Bernal. [8] The C_L of the wing in freestream was measured across a sweep of angle of attack. The wing was then be placed in the gust encounter at a series of set angles of attack. The force history and flow field around the wing was analyzed and compared. The results from the current study will be used to tune a closed loop controller and the wing will be actuated in a closed-loop to mitigate the lift peaks observed during the gust encounter.

II. Experimental Setup

A. University of Dayton Water Tunnel

All experiments were conducted in the University of Dayton Water Tunnel shown in Figure 1. The facility is a horizontal free-surface water tunnel with a 6:1 contraction and a 38 cm wide and 46 cm high test section of 152 cm in length. The speed range of this facility is 0.075 m/s to 0.38 m/s. For this study, all experiments were conducted at a freestream velocity of 0.28 m/s at a Reynolds number of 24,119. The gust generator is operated with an Applied Motion Products HT23-598DC-ZAC stepper motor upstream from the PIV window.

B. Vortical Gust Generator

The gust generator is a flat plate with a span of 18.5 in. and a chord of 3 in. The plate has a thickness of 0.125 inches and runs from a magnetic post at the bottom of the tunnel to the free surface of the tunnel shown in Figure 2b. The gust generator is placed 5 chords upstream from the field of view and the wing mount. It is mounted to the stepper motor and rotates about a shaft magnetically held in place on the bottom of the tunnel. Figure 3b shows the normalized pitch rate (α'_N) is defined in Equation 1 where $\dot{\alpha}$ is the dimensional pitch rate, c is the chord, and U_∞ is the freestream velocity.

$$\alpha'_N = \frac{\dot{\alpha} c}{U_\infty} \quad (1)$$



Figure 1. University of Dayton Water Tunnel Facility

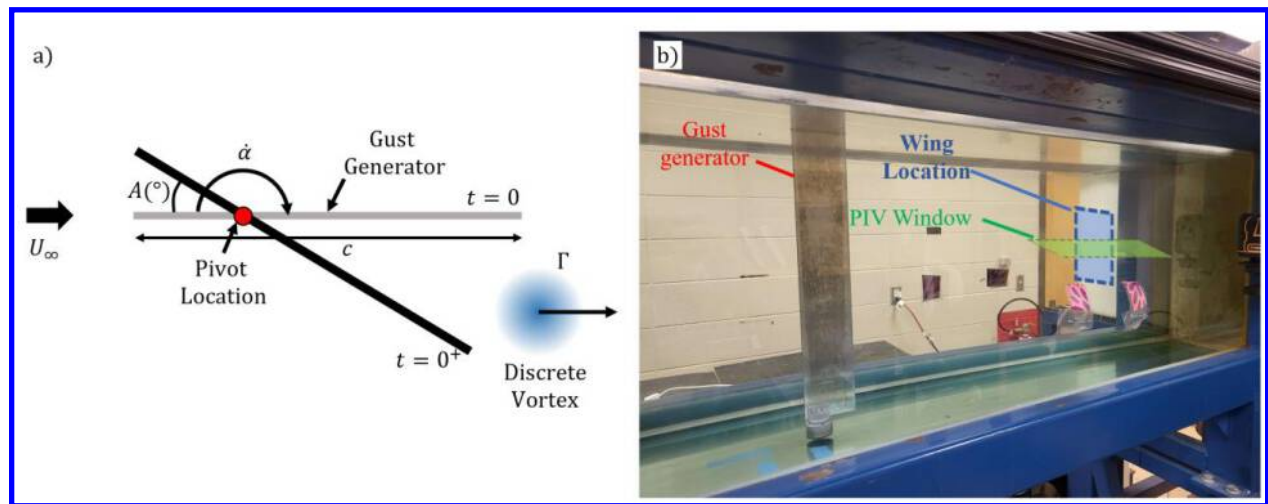


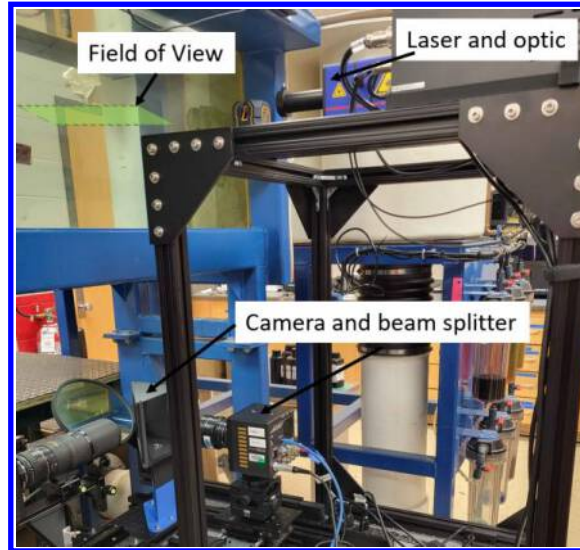
Figure 2. Gust Generator and PIV Window

C. Particle Image Velocimetry

The gust characterization and gust encounters were captured using time resolved PIV. The PIV system used is shown in Figure 3. A FASTCAM Mini highspeed camera was used along with a 532nm LaVision 3-Watt continuous laser for all PIV measurements. The laser beam passed through a Powell lens, spreading it out into a 30-degree fan angle. The horizontal field of view was captured from the glass bottom of the tunnel. The field of view passes through the glass bottom of the tunnel, passes through a 45-degree mirror, and through a beam splitter. The water tunnel was seeded with 60 μ m polyamide particles to be illuminated by the laser sheet. The field of view is 5 chords downstream from the flat plate gust generator and in line with the wing location. Triggering of the camera, releasing the gust, and recording the force and angle of attack data was done with a dSPACE MicroLabBox. The MicroLabBox records the trigger signals which ensures that the PIV frames and the force data can be time synced. Each case was conducted 3 times to average out noise and uncertainty. The PIV frames and the force data was aligned, averaged, and analyzed using MATLAB. The gust profile was tested according to the test matrix in Table 1.

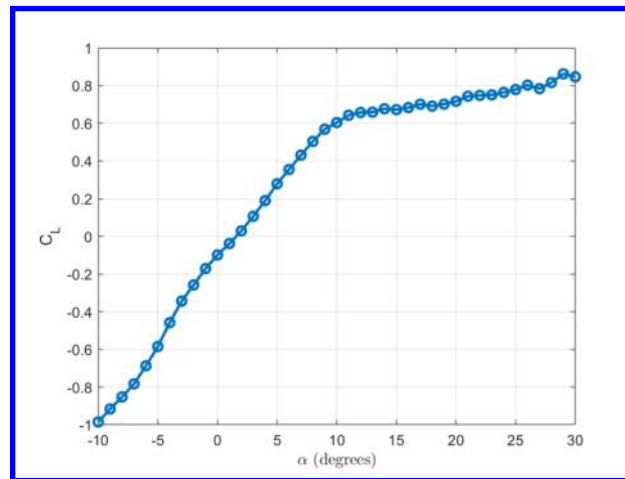
Table 1. Test Matrix for Discrete Vortical Gust Characterization and Encounter

Case	A (°)	$\dot{\alpha}$ (°/s)	α_N	AOA (°)
1	20	165	45	0, 4, 8, 12
2	20	220	60	0, 4, 8, 12

**Figure 3. PIV System in UD Water Tunnel**

D. SD7003 Clear Wing

The SD7003 wing is mounted 5 chords downstream from the gust generator. The wing is attached to a splitter plate resulting in an effective aspect ratio of 4. The Reynolds number for the wing and the gust generator is 24,119. The clear wing will allow the impact of camber on the gust encounter to be investigated using force measurement as well as PIV measurements. It is important to note that there is a small region of occlusion where the laser sheet is distorted due to the camber of the wing. Velocity vectors are unable to be correlated in this region due to the distortion. The distortion appears the most at high angle of attack cases just beneath the leading edge where the camber is greater. The coefficient of lift versus angle of attack can be seen in Figure 4. The SD7003 shows an extended, flat stall which differs from the conventional decrease in C_L seen with other airfoil profiles.

**Figure 4 C_L vs α for SD7003 Wing**

III. Results

A. Discrete Vortex Method Simulation

The ramp-hold-return gust generation profile was modelled using the discrete vortex method. The discrete vortex method (DVM) was used to simulate the generation of a discrete vortex following the same parameters as the experimental tests. The code discretized the upstream gust generator into a series of subpanels which are used to calculate the impact of the gust generator on the surrounding flow. The simulation field was large enough to track the vortex from the gust generator to the downstream experimental field of view. The vortex center was identified by the maximum Q criterion shown in Equation 2. The vortex trajectory was tracked as it travelled downstream to the field of interest. The trajectory of each profile can be seen in Figure 5.

$$Q = \frac{1}{2} (\|\omega\|^2 - \|S\|^2) \quad (2)$$

$$\omega = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \quad (3)$$

$$S = \frac{1}{2} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) \quad (4)$$

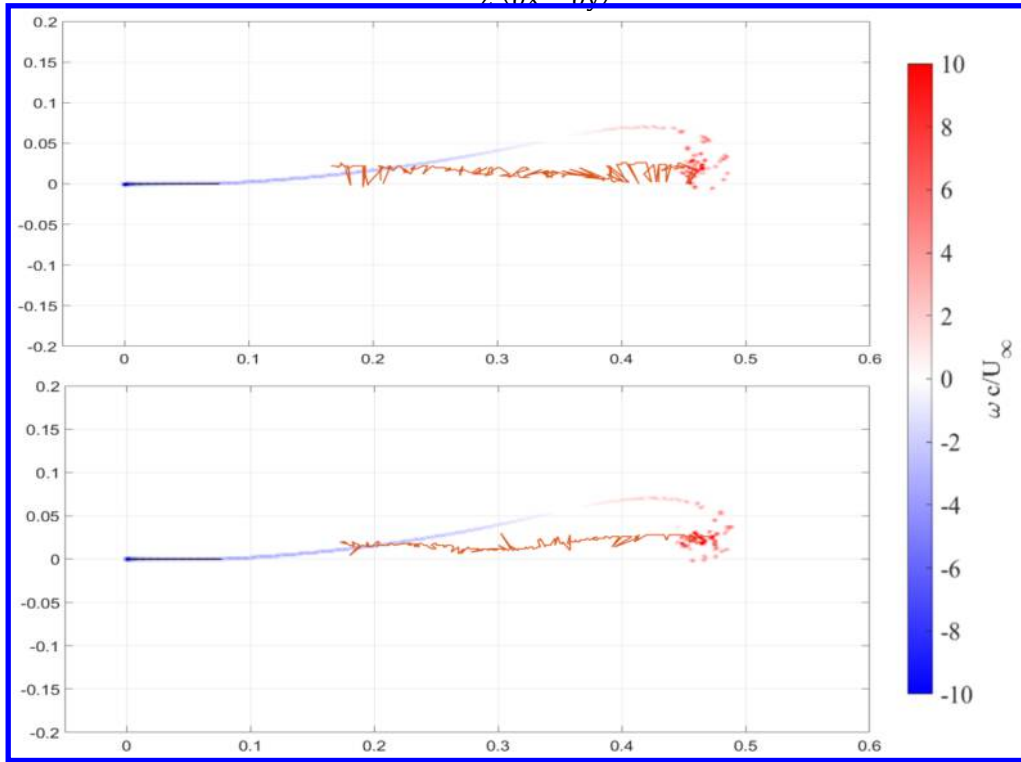


Figure 5 Simulated Trajectories A) $\alpha_N = 45$ B) $\alpha_N = 60$

The normalized circulation was calculated according to Equation 5. The 3D surface plot shown in Figure 6 shows how the circulation grows with an increasing radius. The circulation does not change during the time the vortex is in the area of interest. Since circulation is the area integral of vorticity, the circulation in the surface plot levels off as the radius surpasses the size of the vortex. Once the radius is extended beyond the bounds of the vortex, there is no significant region of vorticity contributing to the circulation integral. The radius associated with the vortex is small due to the lack of viscous effects in the simulation.

$$\Gamma_n = \frac{\Gamma}{U_\infty c} \quad (5)$$

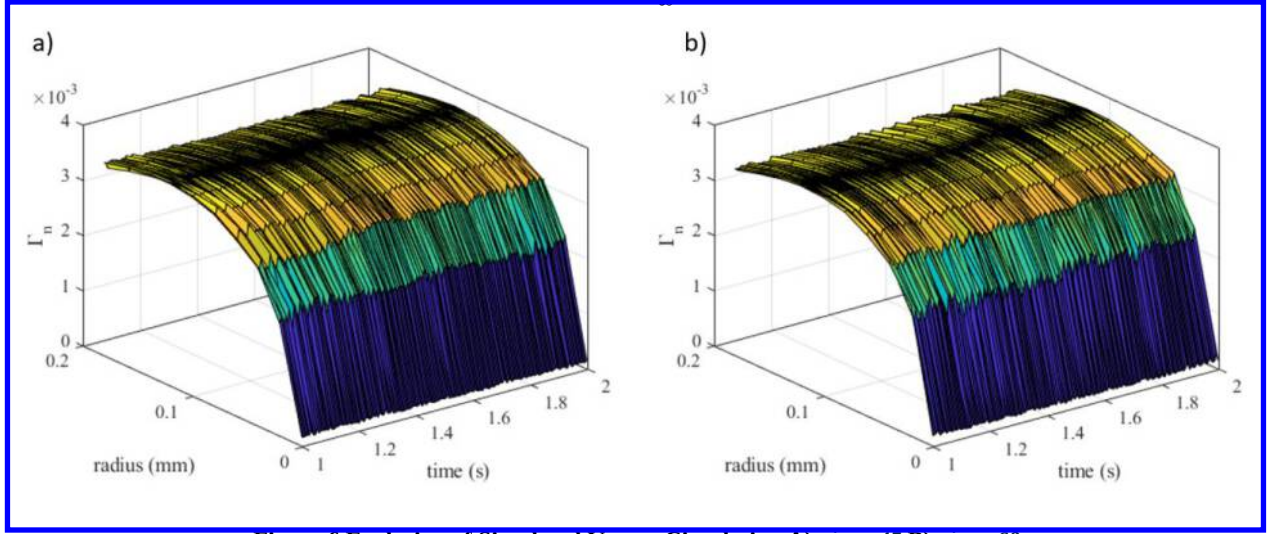


Figure 6 Evolution of Simulated Vortex Circulation A) $\alpha_N = 45$ B) $\alpha_N = 60$

B. Gust Characterization

Coherent vortices of different strength were generated by rapidly pitching a flat plate upstream of the field of view. Figure 7 shows the normalized vorticity spatially unwrapped from each PIV frame. Each profile shows the undisturbed portion of the flow on the right side of the plot. The discrete gust can be seen at the center of the figure, and the wake of the gust generator as it returns to the neutral position is shown on the left. Both gust profiles show a relaxed region after the gust and before the shedding wake of the gust generator begins.

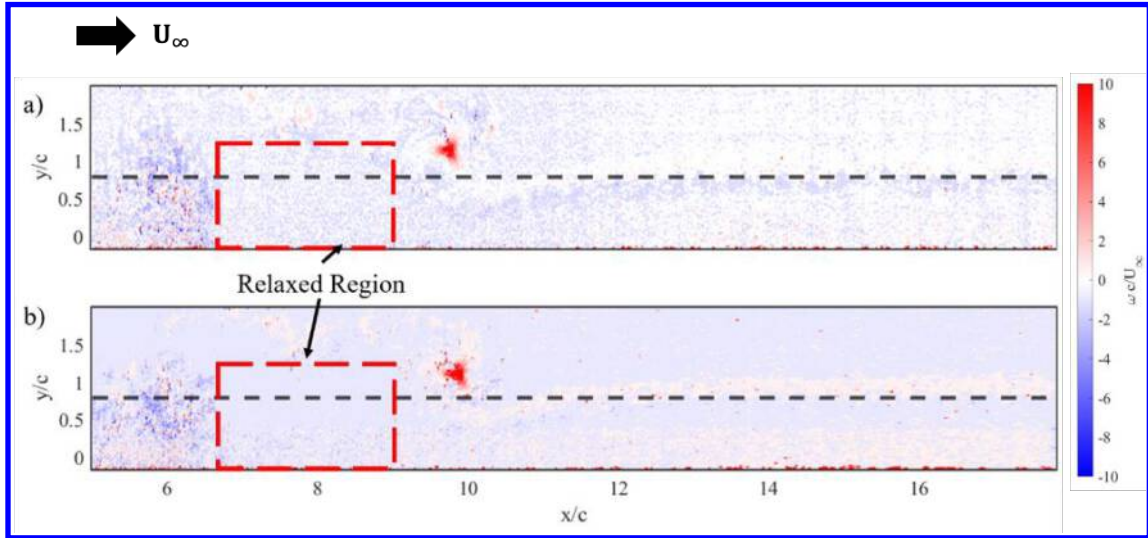


Figure 7. Gust Canvas, a) $\alpha_N = 45$, b) $\alpha_N = 60$

The maximum Q criterion was used to identify the center of the vortex and track the vortex as it passes through the field of view. The deflection of the gust generator offsets the vortex trajectory from the wing location. After the initial deflection, the vortex trajectory through the field of view in Figure 8 is parallel to the wing. The pitch rate of the gust profile does not impact the path of the vortex, but it does impact the coherence of the discrete gust. Normalized pitch rates lower than 45 begin experiencing increased amounts of secondary vortices around the core vortex.

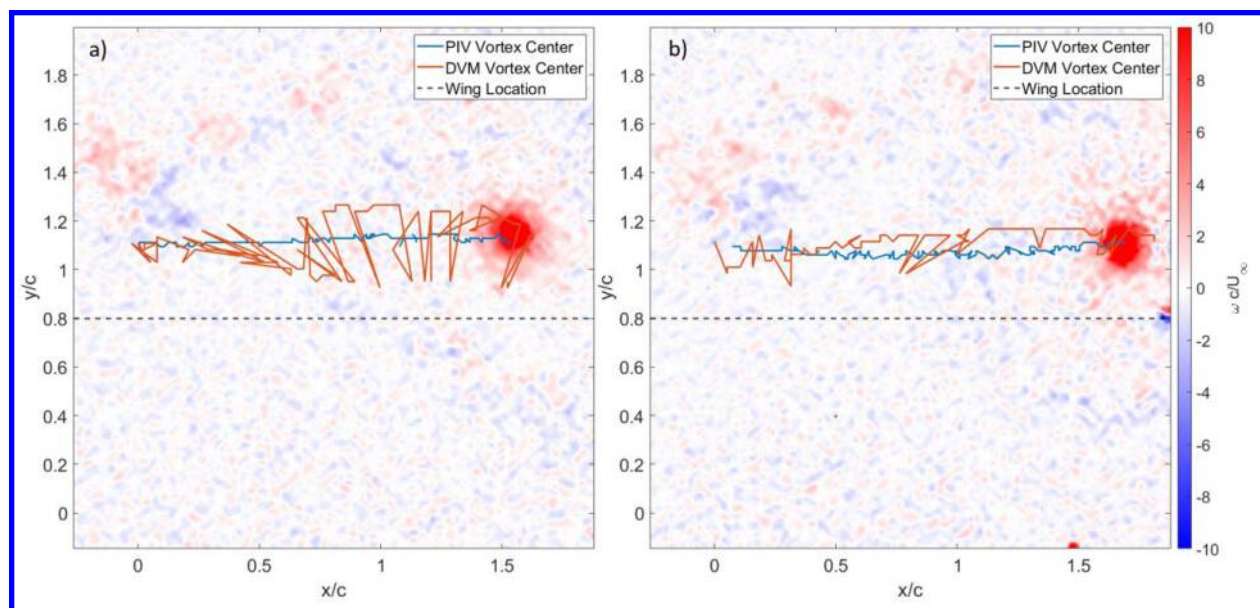


Figure 8. Vortex Trajectories, a) $\alpha_N = 45$, b) $\alpha_N = 60$

A 3D surface plot of the normalized experimental circulation can be seen in Figure 9. The circulation of the experimental gusts does not vary with time as the gust travels through the field of interest. The experimental circulation is significantly higher than in the discrete vortex method simulation. Similarly, the radius at which the circulation levels off is much larger than what was seen in the simulation.

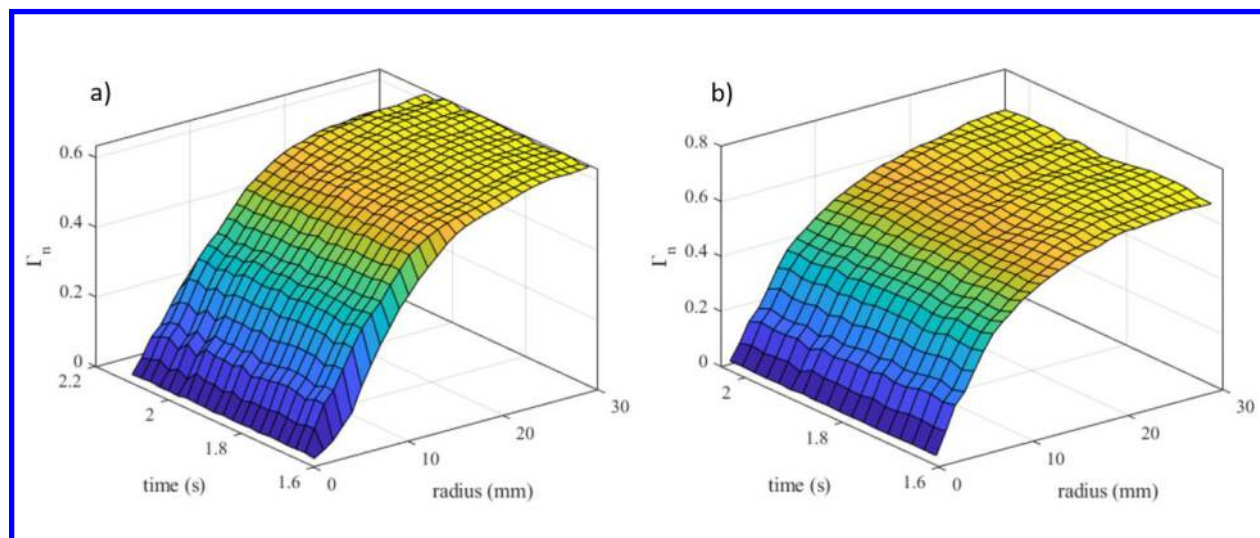


Figure 9. Evolution of Normalized Vortex Circulation for A) $\alpha_N = 45$ B) $\alpha_N = 60$

C. Wing Gust Encounter

A SD7003 wing with a chord of 3 inches was mounted 5 chord lengths downstream of the gust generator. The flow field around the wing as the gust passes through the field of view was quantified with PIV simultaneously with the force history of the encounter. The behavior of the wing's response can be correlated to the vorticity around the wing as the gust passes over the upper surface of the wing. Figure 10 shows the wing's response to the gust profile with a normalized pitch rate of 45. The wing starts out under normal freestream conditions, but as the gust approaches

the leading edge of the wing, the wing experiences a sharp decrease in the coefficient of lift. The frame associated with the negative C_L peak shows a decrease in vorticity on the upper surface of the leading edge of the wing.

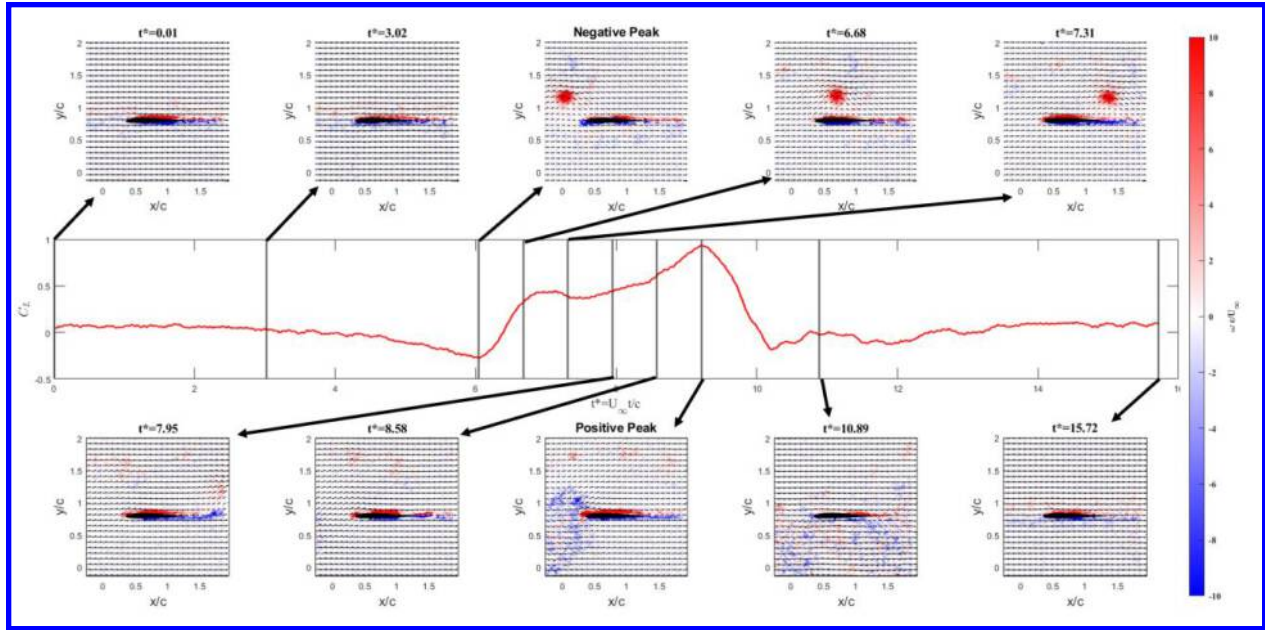


Figure 10. Gust Encounter for $\alpha_N = 45$ and AOA = 0°

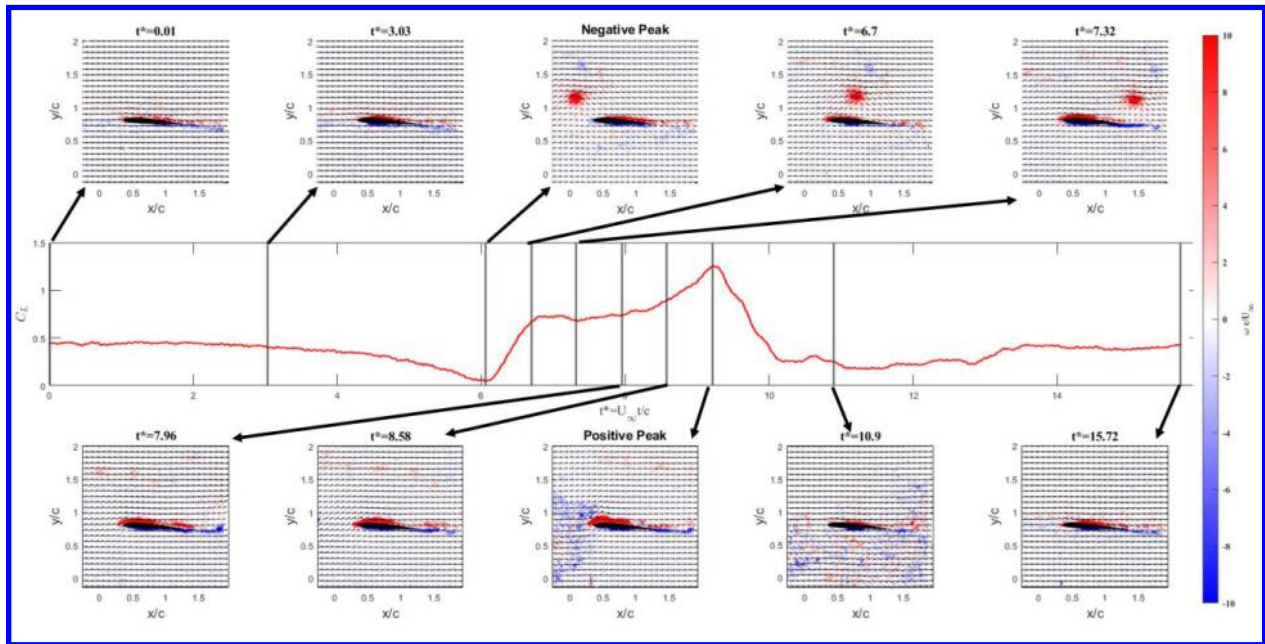


Figure 11. Gust Encounter for $\alpha_N = 45$ and AOA = 4°

The positive vortex is interacting with the positive vorticity on the upper surface of the wing. As the vortex travels along the chord of the wing, the C_L recovers to a greater value than in freestream. Following the initial gust encounter, the C_L increases until it reaches a peak as the wake of the gust generator impinges on the leading edge of the wing. As the wake of the gust generator passes over the wing, the vorticity on the leading edge of the upper surface disappears again, resulting in another decrease in C_L . After the wake of the gust generator passes, the wing returns to pre-gust conditions. The wing-gust encounter when the wing is at 4° is shown in Figure 11. The C_L in this encounter follows

the same trend as the encounter when the wing was at 0° angle of attack. Since the wing shows a higher starting C_L during this encounter, the sharp decrease associated with the gust entry does not cause the wing to experience negative lift. The velocity vectors shown in the frames associated with convective time 7.96 and 8.58 show a positive v velocity component. This creates an induced angle of attack which results in the region of increased C_L between the two peaks. The frame associated with the positive peak in C_L shows a significant increase in vorticity on the upper surface of the leading edge. This occurs as the wake of negative vorticity impinges on the leading edge of the wing.

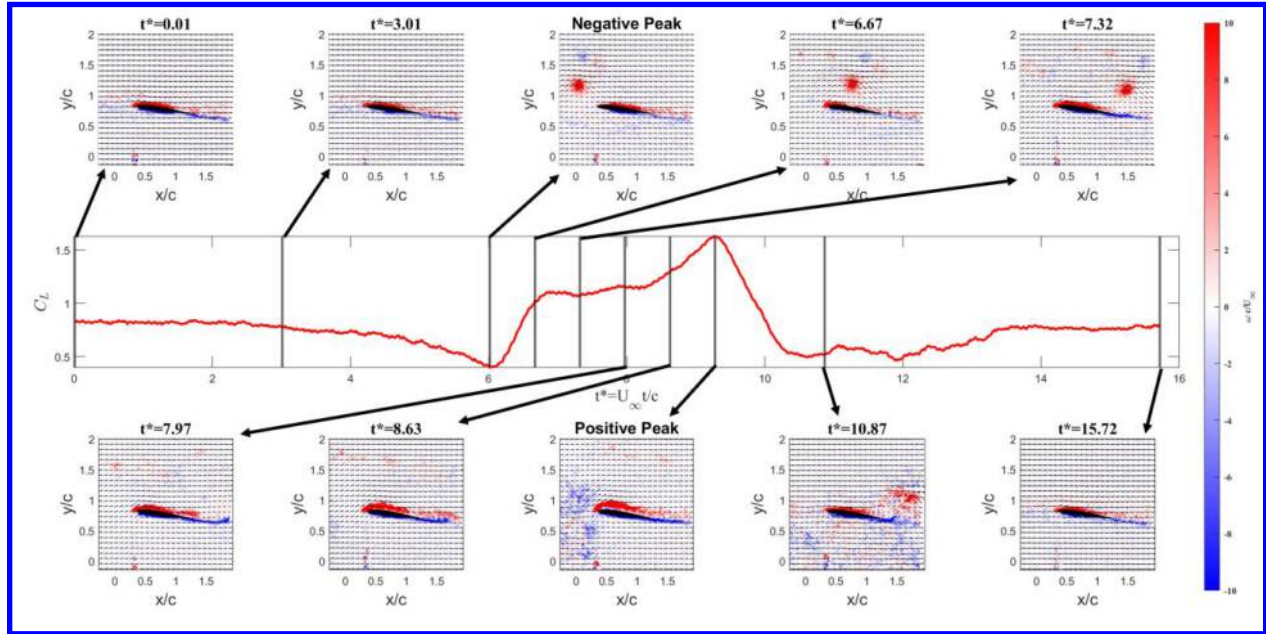


Figure 12. Gust Encounter for $\alpha_N = 45$ and AOA = 8°

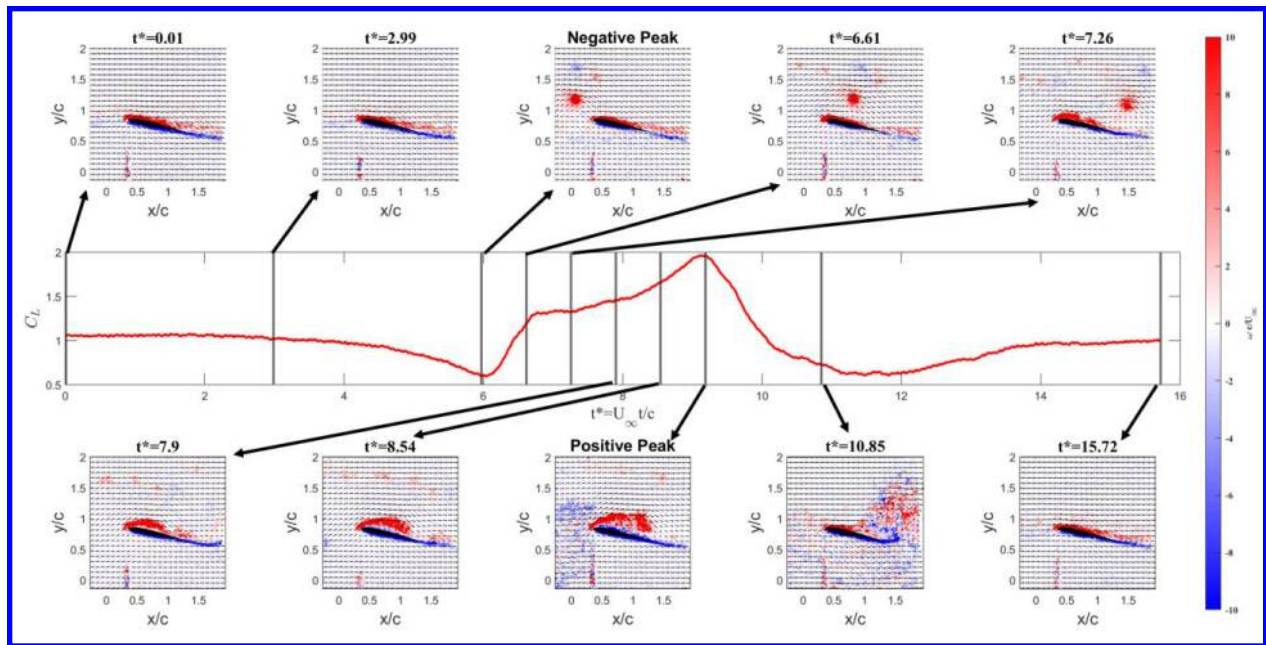


Figure 13. Gust Encounter for $\alpha_N = 45$ and AOA = 12°

The 8-degree case shows the same trend as the 0-degree and 4-degree cases for this gust profile. The frame depicting a convective time of 10.87 in Figure 12 shows the upper surface vorticity after it has shed off the wing. The

vorticity growth in the positive peak frame for the 8-degree case is greater than the previous cases. This vorticity grows as the C_L approaches the positive peak, and the sharp drop in C_L is due to the vorticity separating from the wing and travelling down the chord of the wing. After the flow separation, the wing experiences the remaining wake before returning to freestream conditions. The 12-degree case shown in Figure 13 contains the highest positive C_L peak out of each case. The region between the negative and positive peaks shows strong vorticity along the upper surface of the chord. As the wake of the gust generator impinges on the leading edge of the wing, the upper surface vorticity can be seen beginning to shed off the wing. At a convective time of 10.85, the flow separation can be seen leaving the trailing edge of the wing. The 12-degree case reaches the highest C_L peak and as a result shows the largest vorticity bubble shedding off the upper surface of the wing. It is also important to note that a line of vorticity can be seen below the leading edge of the wing. This is not a physical result, but it is a region of refracted light due to the cambered shape of the wing. The line of vorticity is not a result of the gust profile in this case.

The C_L histories shown in Figure 14 are the encounter from each case with the freestream C_L subtracted from the data. This isolates the fluctuations in C_L due to the gust encounter and removes the effect of the angle of attack. The severity of the negative peak increases with an increase in angle of attack. This is likely due to the proximity of the leading edge as the angle of attack increases. As the angle of attack is increased, the leading-edge pitches up closer to the trajectory of the gust. The proximity between the leading-edge upper surface and the gust increases the influence the gust has on the circulation over the leading edge. The initial recovery at a convective time of 7 shows how angle of attack affects the gust exit. As the gust travels along the chord of the wing, the 0-degree angle of attack case shows how the tail's proximity to the gust influences the C_L . The 0-degree case shows an initial bump in C_L with the 4-degree and 8-degree cases showing a reduced response. Finally, the 12-degree case shows no response as the gust leaves the trailing edge of the wing. In the 12-degree case, the tail of the wing is deflected so far from the gust's trajectory that it sees no significant response.

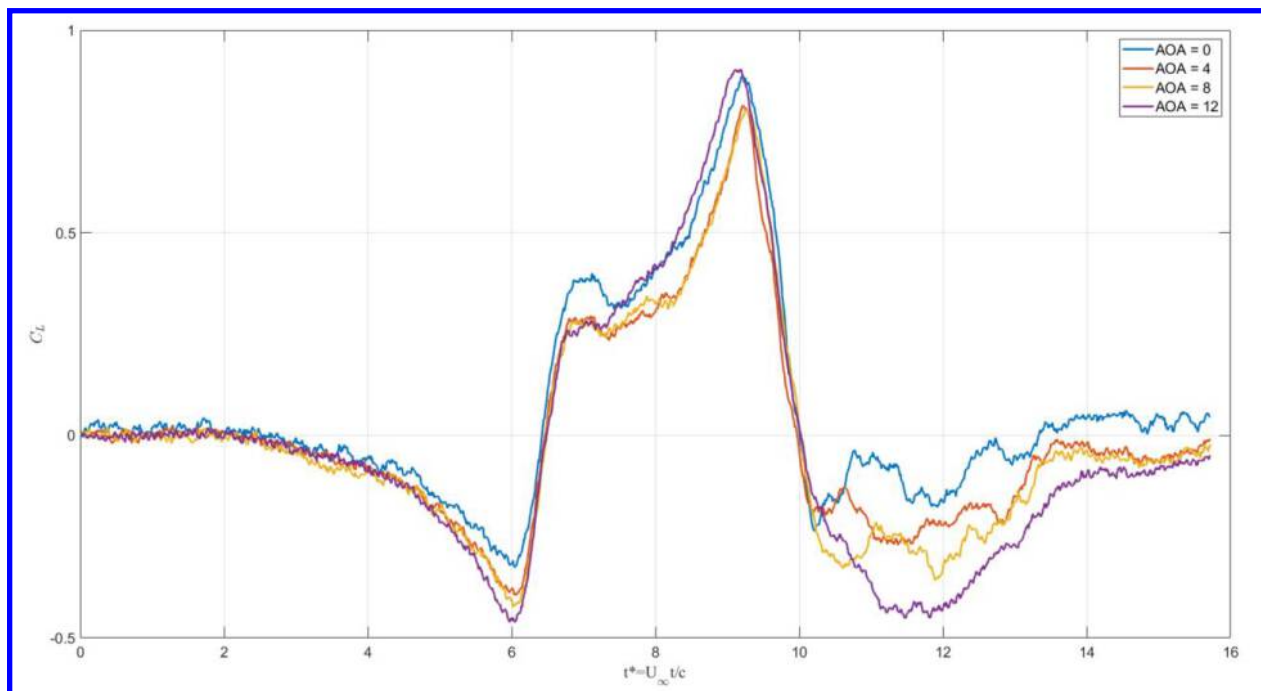


Figure 14 C_L Response at Each AOA

Figure 15 shows the wing's response to the gust profile with a normalized pitch rate of 60 at an angle of attack of 0. Again, the wing experiences a drop-in leading-edge vorticity as the gust enters the field of view. The decrease in vorticity along the upper surface follows the gust as it travels along the chord of the wing. The wing sees a higher than freestream C_L between the negative and positive peaks. The positive C_L peak is followed by the growing vorticity on the upper surface detaching before the wing returns to its freestream behavior.

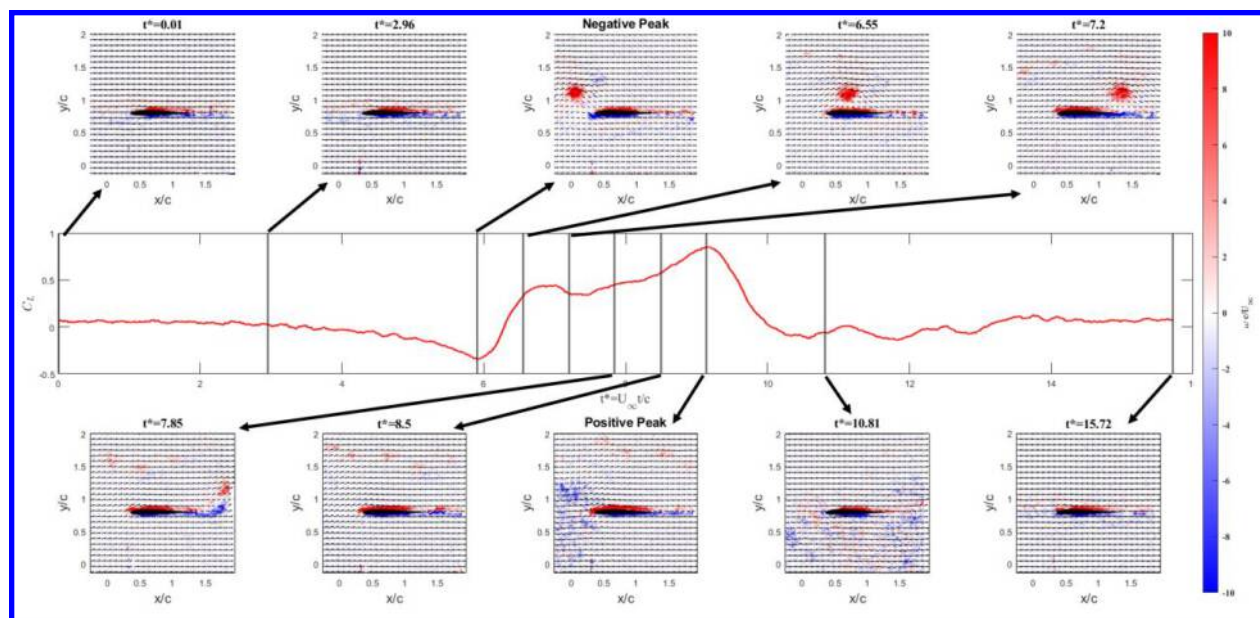


Figure 15 Gust Encounter for $\alpha_N = 60$ and $AOA = 0^\circ$

The 4-degree angle of attack case is shown in Figure 16. The wing begins the encounter with the positive C_L expected from the angle of attack increase. As the gust approaches the leading edge of the wing, the C_L is reduced to 0 at its minimum. The behavior surrounding the bump in C_L during the gust exit holds for the cases with a normalized pitch rate of 60. The 4-degree case shows a reduced recovery bump when compared to the 0-degree angle of attack case.

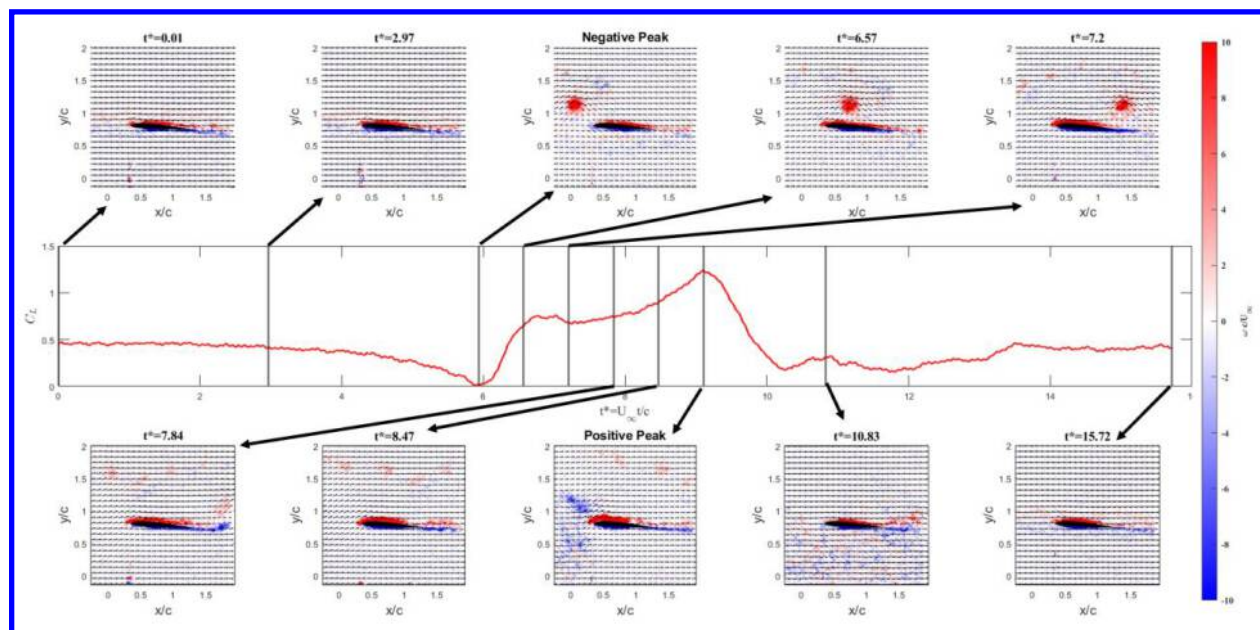


Figure 16. Gust Encounter for $\alpha_N = 60$ and $AOA = 4^\circ$

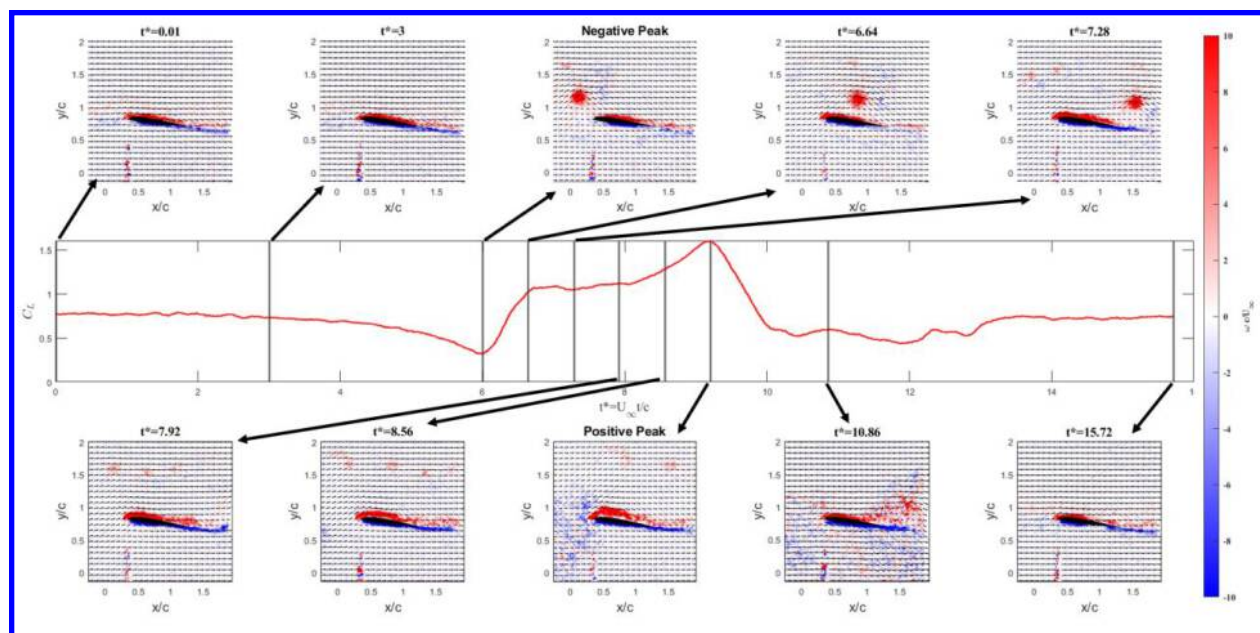


Figure 17. Gust Encounter for $\alpha_N = 60$ and $\text{AOA} = 8^\circ$

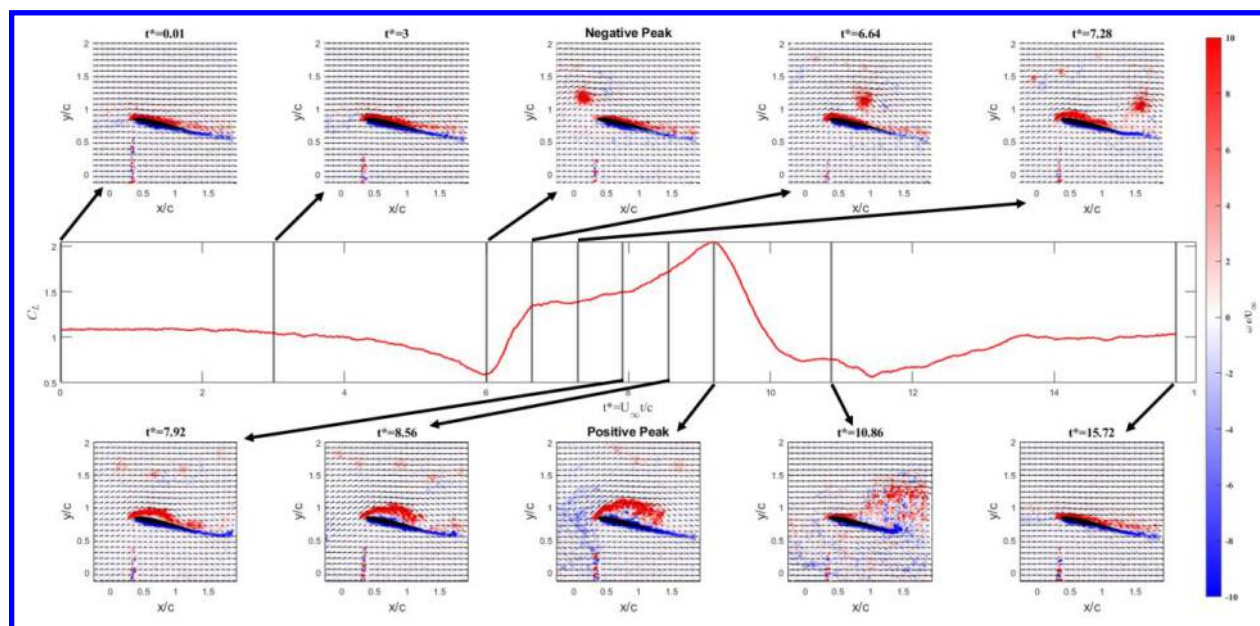


Figure 18. Gust Encounter for $\alpha_N = 60$ and $\text{AOA} = 12^\circ$

The 8-degree case for the normalized pitch rate of 60 is shown in Figure 17. As the convective time moves from 7.28 to the positive peak, the vorticity on the upper surface of the wing increases. This trend continues until the negative vorticity in the wake causes the flow to separate off the upper surface of the wing. The frame associated with a convective time of 10.86 shows the large region of vorticity separating off the upper surface of the wing. The 12-degree angle of attack case shown in Figure 18 has the largest fluctuations of any case. The wing experiences a positive peak C_L over 2. The PIV frame at the positive peak shows strong circulation on the upper surface of the wing. At the convective time 10.86, the flow is fully separated after the mid chord of the wing. The vorticity that grew on the upper surface of the wing can be seen shedding off the wing and out of the field of view. By the time the wake of the gust generator causes the large drop in C_L , the gust generator has returned to the neutral position. The weak wake of the

neutral gust generator can be seen in the first and last frame. The first frame is undisturbed freestream, and the final frame shows the flow has returned to normal conditions.

IV. Conclusions

Discrete vortical gusts were generated by rapidly pitching a flat plate upstream of the field of interest. The size, strength, and trajectory of each gust was simulated using a discrete vortex method and then experimentally calculated using TR-PIV. The impact of the gust on the wing was determined by comparing the force histories of each encounter along with the PIV measurements of the surrounding flow field. The positive vorticity of each gust profile interacted with the vorticity on the upper surface of the leading edge during the gust encounter. The destructive interference decreased the lift experienced by the wing during the gust entry. As the gust passes the trailing edge of the wing and exits the field of view, the wing experienced an increased C_L until the wake of the gust generator caused flow separation on the upper surface. After the wing was exposed to the remaining gust generator wake, the flow relaxed to freestream conditions and the wing experienced C_L equal to pre-gust conditions. The wing's angle of attack impacts the initial C_L recovery due to the differing offsets from the gust trajectory to the trailing edge of the wing. The ramp-hold-return profile succeeds in reducing the length and strength of the gust generator wake. The new profile allows the wing to return to pre-gust conditions. Further tailoring of a ramp-return profile could lead to a discrete gust subsequent shedding wake or counter rotating vortex. A gust encounter without a following wake would allow the C_L recovery region to be investigated without the influence of the negative wake.

Acknowledgments

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